2	(a)	head-on : collision takes place along the line joining the centres of the colliding bodies. elastic collision: both the momentum and the kinetic energy are conserved.	C1 C1
	(b)	speed of separation = speed of approach $v_2 - v_1 = u_1 - 0$ $or\ \ u_1 + v_1 = v_2$ - - - (1)	A1
	(c)	By Conservation of momentum, $mu_1 = mv_1 + 12mv_2$ $\Rightarrow u_1 - v_1 = 12v_2$---(2) $(1)+(2)\ \ 2u_1 = 13v_2 \Rightarrow v_2 = \frac{2}{13}u_1$ from (1) $v_1 = v_2 - u_1 = \frac{2}{13}u_1 - u_1 = -\frac{11}{13}u_1$ Thus ratio $\frac{v_1}{u_1} = \frac{11}{13}$ = 0.846	C1 C1 A1

	(d)	required fraction $= \dfrac{\frac{1}{2}m\left(u_1^{2}-v_1^{2}\right)}{\frac{1}{2}mu_1^{2}} = 1-\left(\dfrac{v_1}{u_1}\right)^2$ $= 1-\left(\dfrac{11}{13}\right)^2 = 0.28$	C1 A1
	(e)	This is a *head-on elastic* collision between two equal masses. After collision, the incident neutron will stop while the target neutron moves off with the velocity of the incident neutron. (The ratio of the final to the initial speed of the incident neutron would be zero.) The fraction of the kinetic energy of the incident neutron that is transferred to the target neutron would be 1.	C1 A1

3(a) The work done by a force on an object particle is defined as the product of the B1